

ATTRIBUTE GRAMMAR FOR IDENTIFICATION FASE

AG = (G, A, R) where

A : expression.definition

G:

(1) Variable: expression $\rightarrow$ ID

(2) FuncDefinition : definition $\rightarrow$ type ID varDefinition* statement*

(3) VarDefinition: definition $\rightarrow$ type ID

R:

Assuming the correct implementation of symboltable

(1)

```
definition = symbolTable.find(ID)
```

```
if definition == null:
```

```
    expression.definition = new VarDefinition(ErrorType("Definition for  
name " + ID + " not found"))
```

```
else
```

```
    expression.definition = definition
```

(2)

```
if !symbolTable.insert(definition):
```

```
    ErrorType("Function " + ID + " already defined")
```

```
symbolTable.set()
```

```
//visit children
```

```
symbolTable.reset()
```

(3)

```
if !symbolTable.insert(definition):
```

```
    ErrorType("Variable " + ID + " already defined")
```